Assignment No 8

Name :- Katkar Abhijit **Div:-** B

Batch:-T3 **Roll No :-** CO3B18

Title – Develop a simple chatbot using predefined rules and pattern matching.

Program:-

```
def chatbot_response(message):
    text = message.lower().strip()

    if any(word in text for word in ["hello", "hi", "hey"]):
        return "Hello! How can I help you?"

    if "bfs" in text:
        return "BFS explores states level by level."

    if "dfs" in text:
        return "DFS explores deeply before backtracking."

    if "a*" in text or "astar" in text:
        return "A* uses f(n) = g(n) + h(n) to guide the search."

    if "viva" in text:
        return "Prepare definitions, algorithms, complexity and code logic."
```

```
    if "lab" in text or "experiment" in text:
        return "Please study the aim, theory, algorithm, code and output."

    return "Sorry, I do not understand that question."



print("Simple Rule-Based Chatbot")
print("Type 'bye' to exit.")

while True:
    message = input("You: ")

    if message.lower().strip() in ["bye", "exit", "quit"]:
        print("Bot: Goodbye! Have a nice day.")
        break

    print("Bot:", chatbot_response(message))
```

Output:-

Simple Rule-Based Chatbot

Type 'bye' to exit.

You: hello

Bot: Hello! How can I help you?

You: bfs

Bot: BFS explores states level by level.

You: a*

Bot: A* uses f(n) = g(n) + h(n) to guide the search.

You: viva

Bot: Prepare definitions, algorithms, complexity and code logic.

You: lab

Bot: Please study the aim, theory, algorithm, code and output.

You: xyz

Bot: Sorry, I do not understand that question.

You: bye

Bot: Goodbye! Have a nice day.

Process finished with exit code 0